Dissipative-particle-based simulation approach for amoeboid cell migration

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Amoeboid cells migrate by coordinating actin-driven protrusion with myosin-driven contraction, allowing rapid shape changes and efficient locomotion in complex environments. Capturing this coupling between intracellular regulation and whole-cell mechanics in a single quantitative framework remains challenging. To quantitatively study amoeboid cell migration, we propose a mesoscopic particle-based model that couples intracellular dynamics with cellular mechanics by combining transport dissipative particle dynamics and smoothed particle hydrodynamics. We represent biochemical regulators as concentration fields carried by individual particles, and compute active forces, such as protrusive and contractile forces, directly from these concentrations. This approach enables us to investigate single-cell migration in complex intra- and extracellular environments, and to connect biochemical patterning to emergent cell shape and motility. The model qualitatively recapitulates experimentally observed phenomena.

I. INTRODUCTION

Amoeboid migration is a fast cell migration mode characterized by low adhesion, relying on cytoplasmic protrusion and cortical contractility. Cells that move in amoeboid mode exhibit fascinating migratory behaviors and dynamics. For example, dendritic cells actively deform when passing through narrow pores [1], and respond to their mechanical environment by optimizing their migration path [2] or adjusting cellular forces [3]. In general, amoeboid motility involves three phases: polarization, protrusion of a leading edge, and contraction of the cell rear (Fig. 1(a)). Polarization is initiated by membrane signaling molecules, such as Rho/Rac proteins, that are activated at the prospective leading edge [4]. These signals trigger actin polymerization, leading to F-actin accumulation at the front and exerting a protrusive force on the plasma membrane, thereby forming pseudopods. Meanwhile, the F-actin filaments flow backward from the leading edge and eventually depolymerize into actin monomers (G-actin), which generates retrograde flow. In adherent (mesenchymal) cells, retrograde flow is partially arrested through connections to the extracellular matrix (ECM) via transmembrane proteins, such as integrins, enabling traction forces to pull the cell forward. By contrast, amoeboid cells do not require strong ECM adhesion and tend to exhibit faster retrograde flow [5]. At the cell rear, myosin motors accumulate and contract the actin network, facilitating retraction of the cell body [6].

Various biophysical models have been developed in recent decades to study amoeboid migration. At the continuum level, active gel theory has been used to describe contractility-driven amoeboid motility in one dimension [7], and extended to incorporate cell morphodynamics under confinement in a minimal hydrodynamic framework [8]. A Stokes-flow continuum model with dynamically discretized cortex and nucleus contours has also been proposed for migration in complex ECM [9]. Phase-field models offer flexibility in handling deformable boundaries and have been employed to describe cell shape evolution [10], the coupling between cytosolic dynamics and membrane mechanics [11, 12], and the integration of signaling pathways with motility [13]. These models typically solve coupled partial differential equations for continuum fields. Beyond continuum descriptions, discrete models can also capture essential features of amoeboid motion. In particular, bleb-driven motility mediated by membrane–cortex interactions has been described by a vertex-based modeling approach [14]. A vertex-based method has also been employed for cortical-flow-driven single-cell and collective migration [15]. Moreover, cellular Potts models have been proposed for multicompartment cells migrating under confinement [16, 17], without explicitly describing intracellular dynamics. In contrast, the cellular Potts model incorporated with local actin dynamics has also been considered. [18].

Although some modeling frameworks, such as phase-field and active-gel methods, have successfully integrated cytoskeletal dynamics with cell-level motion, coarse-grained particle-based approaches remain relatively underexplored. A particle-based description is particularly advantageous for cell migration in environments with sophisticated geometry, where continuum discretizations may become cumbersome and can suffer from numerical instabilities. Moreover, its bottom-up construction

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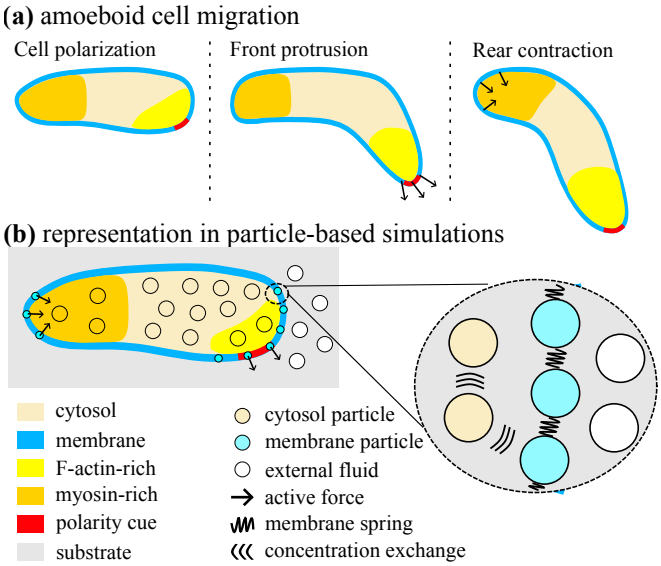

FIG. 1. Schematic depiction of amoeboid motion and the model. (a) Three stages in amoeboid motion: polarization, protrusion, and contraction. (b) Model configuration. The simulation model comprises particles carrying values of biochemical concentration fields that regulate the cell migration. The cell consists of a spring-connected particle chain and cytosolic particles. Cell polarity is determined stochastically on the membrane particles. F-actin polymerization is then triggered in the vicinity of the polarity particles, while myosin II dynamics occurs throughout the cell. Active forces, i.e., protrusive and contractile forces, are computed from the corresponding concentration fields carried by particles and exerted on the membrane particles.

relies on local interactions between neighboring particles, which naturally accommodates microenvironmental heterogeneity and makes it straightforward to incorporate additional subcellular processes without introducing nonlocal interactions at the continuum level. To bridge the gap, we introduce a mesoscopic model based on the transport dissipative particle dynamics (tDPD) [19] and smoothed particle hydrodynamics (SPH) [20] to couple intracellular dynamics with cell mechanics. Table I presents qualitative distinctions between our particle-based model and continuum models, i.e., phase-field models. Our approach provides a coarse-grained, bottom-up modeling framework for amoeboid migration in complex fluidic environments.

In this model, the cell boundary is represented by a deformable particle-based layer that coarse-grains the membrane–cortex composite. We examine our model for amoeboid migration on a two-dimensional surface and a confined environment micropatterned by pillars, demonstrating behaviors consistent with published theories and experiments.

II. THE MODEL

A. Model description

In our model, molecules involved in cellular dynamics are represented by biochemical concentration fields carried by particles. The tDPD framework enables the simulation of intra- and extracellular fluid flows, along with advection-diffusion-reaction (ADR) of these concentration fields. The cell membrane is modeled by a chain of spring-connected particles. Active forces, such as protrusive and contractile forces, are computed from the relevant concentration fields using an SPH scheme, thereby coupling biochemical regulation and cellular mechanics. As illustrated in Fig. 1, we use three different particle species to represent the cytosol, the membrane, and the external fluid, respectively. To describe intracellular dynamics, we use four concentration fields: F-actin, G-actin, cortex-recruitable myosin, and cytosolic myosin. Each particle carries values of all four concentration fields. Concentration exchange is restricted within the cell, namely between cytosol–cytosol, cytosol–membrane, and membrane–membrane particle pairs, while exchanges between cellular and external fluid particles do not occur. This restriction enforces no-flux boundary conditions for biochemical species at the cell membrane. Furthermore, membrane particles form a spring-connected, deformable boundary that mechanically separates the cytosol from the external fluid and prevents leakage of particles from the interior of the cell. Cell polarization is modeled as a stochastic process: at each timestep, a Bernoulli trial selects a membrane particle as a polarization site, where actin polymerization occurs preferentially, with a prescribed probability. Active forces arising from local biochemical concentrations are then computed for the affected particles and applied directly as body forces in the hydrodynamic equations, analogous to Navier–Stokes with external forcing. The model is implemented as a customized software package within the framework of the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [21].

B. General framework

We consider a system of interacting tDPD particles indexed by i, each characterized by its position $\mathbf{r}_i$, mass m_i, and a set of carried concentrations C_i. The governing equations for tDPD interactions are given by

$$m_i \frac{\mathrm{d}^2\mathbf{r}_i}{\mathrm{d}t^2} = \sum_{j\neq i} (\mathbf{F}^C_{ij} + \mathbf{F}^D_{ij} + \mathbf{F}^R_{ij}) + \mathbf{F}^A_i + \mathbf{f}_i \tag{1}$$

$$\frac{\mathrm{d}C_i}{\mathrm{d}t} = \sum_{j\neq i} (Q^D_{ij} + Q^R_{ij}) + Q^S_i. \tag{2}$$

TABLE I. Qualitative comparison of cell migration.

Feature	Particle-based model (this work)	Continuum model	Other discrete model
Approach	tDPD+SPH	phase-field/active-gel	Vertex/CPM[a]
Discretization	particles	mesh	vertices/lattice sites
Cellular dynamics	pairwise interaction	coupled PDEs	energy-based
Cell boundary	explicit membrane	diffuse interface	polygon/lattice
Geometry	obstacle particles	mesh domain	vertices/lattice
No-flux boundary	via disable flux exchange	boundary conditions	rule-based constraints [a]

[a] CPM refers to cellular Potts Models.

The pairwise interaction forces $\mathbf{F}_{ij}^C$, $\mathbf{F}_{ij}^D$, and $\mathbf{F}_{ij}^R$ are the conservative, dissipative, and random DPD forces between particles i and j, respectively [22]. $\mathbf{F}_i^A$ is an active force arising from intracellular regulation, and $\mathbf{f}_i$ denotes substrate friction. The concentration dynamics encoded by Eq. (2) comprise a deterministic (Fickian) exchange term Q_{ij}^D, a stochastic exchange term Q_{ij}^R between particle pairs (i, j), and a local source term Q_i^S accounting for chemical reactions [19]. Explicit functional forms of these forces and fluxes are specified in later sections.

For the momentum balance, the pairwise forces take the conventional forms,

$$\mathbf{F}_{ij}^C = a_{ij}\omega_C(r_{ij})\mathbf{e}_{ij}, \tag{3}$$

$$\mathbf{F}_{ij}^D = -\gamma_{ij}\omega_D(r_{ij})(\mathbf{e}_{ij}\cdot\mathbf{v}_{ij})\mathbf{e}_{ij}, \tag{4}$$

$$\mathbf{F}_{ij}^R = \sigma_{ij}\omega_R(r_{ij})\xi_{ij}(\Delta t)^{-1/2}\mathbf{e}_{ij}, \tag{5}$$

where a_{ij} is the repulsive coefficient, $\mathbf{e}_{ij}$ is the unit vector defined as $\mathbf{r}_{ij}/|\mathbf{r}_{ij}|$, $\mathbf{v}_{ij}$ is the relative velocity $\mathbf{v}_i - \mathbf{v}_j$ with $\mathbf{v}_j = \mathrm{d}\mathbf{r}_j/\mathrm{d}t$, γ_{ij} is the dissipative coefficient, and σ_{ij} is the amplitude of the random force for the particle pair (i, j). The normalized weighting functions, ω_C, ω_D, and ω_R, determine the distance dependence of interactions within the cutoff radius. ξ_{ij} is a Gaussian random number with zero mean and unit variance. The active force $\mathbf{F}_i^A$ arises from signaling pathways and is determined by specific concentration fields.

For the concentration fields, the pairwise concentration fluxes are

$$Q_{ij}^D = -\kappa_{ij}\omega_{DC}(r_{ij})(C_i - C_j), \tag{6}$$

$$Q_{ij}^R = \epsilon_{ij}(C_i + C_j)\omega_{RC}(r_{ij})(\Delta t)^{-1/2}\zeta_{ij}, \tag{7}$$

where κ_{ij} and ϵ_{ij} control the strength of the fluxes, ω_{DC} and ω_{RC} are weighting functions, and ζ_{ij} is a Gaussian random number with zero mean and unit variance. The fluctuation-dissipation theorem requires [19, 23]

$$\sigma_{ij}^2 = 2k_BT\gamma_{ij}, \quad \omega_D(r_{ij}) = \omega_R^2(r_{ij}), \tag{8}$$

$$\epsilon_{ij}^2 = m_s^2\rho\kappa_{ij}, \quad \omega_{DC}(r_{ij}) = \omega_{RC}^2(r_{ij}), \tag{9}$$

where m_s is the mass of the solute molecules and ρ is the number density of tDPD particles. For simplicity, we use a linear weighting function

$$\omega(r; r_0) = \begin{cases} 1 - \dfrac{r}{r_0}, & r \le r_0, \\ 0, & r > r_0, \end{cases} \tag{10}$$

and set

$$\omega_C(r_{ij}) = \omega_R(r_{ij}) = \omega(r_{ij}; r_c), \quad \omega_D(r_{ij}) = \omega^2(r_{ij}; r_c), \tag{11}$$

$$\omega_{RC}(r_{ij}) = \omega(r_{ij}; r_{cc}), \quad \omega_{DC}(r_{ij}) = \omega^2(r_{ij}; r_{cc}), \tag{12}$$

where r_c and r_{cc} are the cutoffs for DPD interactions and concentration exchange, respectively.

C. Cell membrane

We model the two-dimensional cell boundary as a closed chain of membrane particles connected by springs. The mechanical energy of the boundary is

$$E_{\text{shape}} = E_s + E_P + E_{\text{bending}} + E_{\text{spring}}, \tag{13}$$

where E_S, E_P, E_{bending}, and E_{spring} account for area elasticity, perimeter elasticity, bending rigidity, and spring elasticity, respectively. Specifically, we use

$$E_S = K_S\,(S - S_0)^2, \tag{14}$$

$$E_P = K_P\,(P - P_0)^2, \tag{15}$$

$$E_{\text{bending}} = \sum_i K_\theta\,(\theta_i - \theta_0)^2, \tag{16}$$

$$E_{\text{spring}} = \sum_i K_l\,(l_i - l_0)^2, \tag{17}$$

where S and P are the instantaneous enclosed area and perimeter, and S_0 and P_0 are the target values. The local turning angle at particle i is denoted by θ_i with rest angle θ_0, and l_i is the length of membrane edge i with rest length l_0. The coefficients K_S, K_P, K_θ, and K_l are the corresponding elastic constants. The explicit forms of these terms follow Ref. [24]. The resulting mechanical force on membrane particle i is obtained from the negative of the energy gradient,

$$\mathbf{F}_{i,\text{shape}} = -\nabla_i E_{\text{shape}}. \tag{18}$$

For clarity, throughout the manuscript, the terms "membrane" and "cortex" refer to the coarse-grained boundary representation.

D. Cell polarity

We focus on the regime where the establishment of cell polarity is assumed to be much faster than the timescales of pseudopod growth and migration, although experimental studies indicate that in some contexts polarity dynamics can evolve on comparable timescales [25]. Therefore, we simplify the formation of polarity as a stochastic process, where a membrane particle $\mathbf{r}_s$ is selected as a reference site for a new pseudopod by local F-actin polymerization. We define a polymerization radius r_{poly}, such that polymerization occurs within the cytosolic region $|\mathbf{r} - \mathbf{r}_s| < r_{\text{poly}}$.

To determine the reference particle $\mathbf{r}_s$, we assume that in the absence of environmental bias, the cell follows a persistent random walk [26], which becomes diffusive over long timescales. We then model pseudopod dynamics as a Poisson process that acts on membrane particles. A pseudopod is assigned a lifetime τ_{live}, which yields a probability of death $P_{\text{death}} = 1 - \exp(-\Delta t/\tau_{\text{live}})$ per timestep Δt. The probability of activating a new protrusive site is characterized by the birth rate α_{birth}, whereby $P_{\text{birth}} = 1 - \exp(-\alpha_{\text{birth}}\Delta t)$. We denote the protrusive sites generated by this process as $\mathbf{r}_{\text{prot}}$.

Short-timescale fluctuations around the protrusive direction are then described by an Ornstein–Uhlenbeck (OU) process [27].

For a protrusive direction $\mathbf{r}_{\text{prot}}$, we update $\mathbf{r}_s$ by

$$\mathbf{r}_s(t+\Delta t) = \mathbf{r}_{\text{prot}} + e^{-\Delta t/\tau}\big(\mathbf{r}_s(t) - \mathbf{r}_{\text{prot}}\big) + \sqrt{\sigma\tau\big(1 - e^{-2\Delta t/\tau}\big)}\,\xi, \tag{19}$$

where τ is the relaxation time, σ the noise strength, and $\xi \sim \mathcal{N}(0,1)$. Equation (19) is evaluated along the membrane contour via arc-length index.

E. Actin dynamics

The cytoskeletal dynamics are modeled with actin concentration fields consisting of two constituents: F-actin, which contributes to the protrusive force, and G-actin monomers, which are substrates for forming F-actin, and we assume that the total mass of actin monomers is conserved inside the cell. For simplicity, we consider that the F-actin polymerization is triggered when G-actin monomers come close to each other with a variable polymerization coefficient $P(C_{i,f}, C_{i,g})$, leading to the following expression,

$$Q^S_{i,f} = \begin{cases} P_i(C_{i,f}, C_{i,g})C^2_{i,g} - \mu_f C_{i,f}, & |\mathbf{r}_i - \mathbf{r}_s| \le r_{\text{poly}} \\ -\mu_f C_{i,f}, \end{cases} \tag{20}$$

$$Q^S_{i,g} = -Q^S_{i,f}, \tag{21}$$

where the subscripts f and g denote F-actin and G-actin, respectively. $P(C_{i,f}, C_{i,g})$ is the polymerization rate-coefficient depending on the local concentrations of F-actin and G-actin, μ_f is the depolymerization rate of F-actin, and $Q^S_{i,f} + Q^S_{i,g} = 0$ guarantees that the conversion between F-actin and G-actin conserves the total actin mass locally at each tDPD particle. Thus, no net monomers are created or destroyed by the reaction terms [28]. The polymerization coefficient is given by

$$P_i(C_{i,f}, C_{i,g}) = p\frac{C^{n_g}_{i,g}}{C^{n_g}_{i,g} + K^{n_g}_g}\frac{K^{n_f}_f}{K^{n_f}_f + C^{n_f}_{i,f}}, \tag{22}$$

where p sets the maximum polymerization rate, and the two Hill functions control the saturation by G-actin and inhibition by F-actin with the half-saturation parameter K_g, half-inhibition parameter K_f, and the corresponding Hill coefficients n_g, n_f. Throughout our simulation, we choose $n_g = n_f = 2$. Qualitatively, Eqs. (20) and (22) describe a self-limiting polymerization process, where accumulation of F-actin reduces further polymerization.

F. Myosin dynamics

Motivated by the existence of distinct cytosolic and cytoskeletal myosin II populations [29, 30], we represent myosin with two interconverting fields, a cortex-recruitable myosin C_{m_a}, and a soluble pool C_{m_b}. For simplicity, we adopt a similar functional form of the source terms of actin to describe the conversion, which reads

$$Q^S_{i,m_a} = A_i(C_{i,m_b}, C_{i,f})C^2_{i,m_b} - \mu_m C_{i,m_a}, \tag{23}$$

$$Q^S_{i,m_b} = -Q^S_{i,m_a}. \tag{24}$$

To account for the spatial preference of myosin in cells migrating in amoeboid mode [3, 6, 30], we introduce a conversion coefficient A_i, which uses the F-actin field $C_{i,f}$ as the proxy of cell polarity, to describe myosin redistribution phenomenologically. We express A_i as

$$A_i(C_{i,m_b}, C_{i,f}) = A_{\max}\frac{C^{n_{m_a}}_{i,m_b}}{C^{n_{m_a}}_{i,m_b} + K^{n_{m_a}}_{m_b}}\frac{K^{n_{m_b}}_{f,m}}{K^{n_{m_b}}_{f,m} + C^{n_{m_b}}_{i,f}}, \tag{25}$$

where $A_{\max}$ is the maximum conversion rate. Hill coefficients n_{m_a} and n_{m_b} are set to $n_{m_a} = n_{m_b} = 2$. The parameters K_{m_b} and $K_{f,m}$ set the half-saturation for the soluble-to-cortex-recruitable conversion and the strength of F-actin-mediated suppression, respectively. Importantly, the $C_{i,f}$-dependent factor is not meant to imply that F-actin biochemically inhibits C_{i,m_a}; rather, it is a phenomenological polarity coupling that biases cortex-recruitable myosin toward the rear to support contractility.

To clarify how the above two-pool description relates to commonly used continuum models for myosin transport [11, 31], we consider the dynamics of the total myosin concentration at the i-th particle $C_{i,m} = C_{i,m_a} + C_{i,m_b}$. Since the source terms are antisymmetric, shown by Eq. (24), reactions only redistribute myosin between pools and do not induce a net change of the total myosin.

Therefore, the evolution of $C_{i,m}$ is determined solely by transport through pairwise exchange with neighboring particles j within the interaction range. Summing the transport contributions of the two pools yields

$$\frac{\mathrm{d}C_{i,m}}{\mathrm{d}t} = \sum_j -\omega^2(r_{ij})\Big(\kappa_{\mathrm{eff},i}C_{m,i} - \kappa_{\mathrm{eff},j}C_{m,j}\Big), \quad (26)$$

where the effective mobility $\kappa_{\mathrm{eff},i}$ is given by $\kappa_{\mathrm{eff},i} = \kappa_{m_a}\rho_i + \kappa_{m_b}(1-\rho_i)$. κ_{m_a} and κ_{m_b} are the pairwise flux exchange parameters for cortex-recruitable and cytosolic myosin, respectively. ρ_i is the mass fraction $\rho_i = C_{i,m_a}/C_{i,m}$. Moreover, we assume that myosin is dilute in the cytosol; therefore, we neglect the random fluxes in Eq. (26) due to the fluctuation-dissipation relation for tDPD particles [19]. The continuum limit form of Eq. (26) reads

$$\frac{\partial C_m}{\partial t} + \mathbf{u}\cdot\nabla C_m = \nabla^2\big(D_{\mathrm{eff}}(\rho)C_m\big), \quad (27)$$

where $\mathbf{u}$ is the flow field, $D_{\mathrm{eff}}(\rho)$ is the effective diffusivity obtained by kernel integration of $\kappa_{\mathrm{eff},j}$ over neighbors [19] and it is a function of the mass fraction $\rho = C_{m_a}/C_m$. In our two-pool description, the polarity proxy C_f regulates the mass fraction ρ, which in turn sets the effective transport coefficient D_{eff} through $\kappa_{\mathrm{eff}} = \kappa_{m_a}\rho + \kappa_{m_b}(1-\rho)$. We assume that the cortex-recruitable myosin C_{m_a} is less diffusive than the cytosolic myosin C_{m_b} due to the potential binding, $\kappa_{m_a} < \kappa_{m_b}$. As a result, D_{eff} becomes spatially graded along the front–rear axis, i.e., D_{eff} is higher at the cell front where C_f is rich. Writing the total flux of Eq. (27),

$$\mathbf{J} = C_m\mathbf{u} - \nabla(D_{\mathrm{eff}}C_m) = C_m\mathbf{u} - D_{\mathrm{eff}}\nabla C_m - C_m\nabla D_{\mathrm{eff}},$$

shows that rear enrichment of total myosin is biased by the mobility-gradient flux $-C_m\nabla D_{\mathrm{eff}}$, which drives total myosin down the gradient of D_{eff}, namely to the cell rear, while the Fick-like term $-D_{\mathrm{eff}}\nabla C_m$ relaxes concentration gradients. For brevity, we refer to the cortex-recruitable myosin pool C_{m_a} as "myosin" for the presentation of results, unless explicitly stated otherwise.

G. Active forces

The active forces consist of the protrusive force from the F-actin field and the contractile force from the myosin field. To couple the concentration fields to the fluidic body forces, we adopt SPH [20] in our model. In SPH, the gradient of a scalar concentration field c at particle i is approximated by

$$\begin{aligned}\nabla c_i &= \sum_j \frac{m_j}{\rho_j}(c_j - c_i)\,\nabla W(r_{ij}) \\ &= \sum_j \frac{m_j}{\rho_j}(c_j - c_i)\,W'(r_{ij})\,\frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}},\end{aligned} \quad (28)$$

where the kernel function $W(r)$ and its derivative $W'(r)$ follows

$$W(r) = \Big(1 - \frac{r}{r_c}\Big)^n \quad \text{for } r \le r_c, \quad (29)$$

$$W'(r) = -\frac{n}{r_c}\Big(1 - \frac{r}{r_c}\Big)^{n-1} \quad \text{for } r \le r_c, \quad (30)$$

and vanishes for $r > r_c$. Here $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, $n = 2$ is the kernel exponent, and r_c is the SPH cutoff radius, m_i is the mass of the i-th particle, and ρ_i is the mass density of the i-th particle. We assume constant per-particle mass, and the incompressibility maintains a nearly uniform particle number density. Therefore, we can rewrite the above gradient as

$$\nabla c_i = \frac{1}{V_0}\sum_j (c_j - c_i)W'(r_{ij})\frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}, \quad (31)$$

where V_0 is the per-particle volume. For the F-actin field, we employ a protrusive stress $\sigma_{i,f}$ [11],

$$\sigma_{i,f} = -\frac{\zeta_f C_{i,f}}{|\nabla C_{i,f}|^2}\nabla C_{i,f}\otimes\nabla C_{i,f}, \quad (32)$$

where $\otimes$ is the outer product, with a protrusion coefficient ζ_f, and $\nabla C_{i,f}$ takes the form of Eq. (31). The protrusive force is then given by

$$\begin{aligned}\mathbf{F}_{i,f} &= -\nabla\cdot\sigma_{i,f} \\ &= \frac{1}{V_0}\sum_j(\sigma_{i,f} - \sigma_{j,f})W'(r_{ij})\frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}.\end{aligned} \quad (33)$$

An isotropic contractile stress [7, 11] for the myosin field is expressed as

$$\sigma_{i,m} = \zeta_m C_{i,m_a}\mathbf{I}, \quad (34)$$

where $\mathbf{I}$ is the identity matrix, ζ_m is the contractile coefficient. Then, the contractile force reads

$$\mathbf{F}_{i,m} = -\nabla\cdot\sigma_{i,m} = -\zeta_m\nabla C_{i,m_a}, \quad (35)$$

where $\nabla C_{i,m_a}$ is calculated by Eq. (31). The total active force $\mathbf{F}_i^A$ in Eq. (1) is given by

$$\mathbf{F}_i^A = \mathbf{F}_{i,f} + \mathbf{F}_{i,m}. \quad (36)$$

For simplicity, we set $V_0 = 1$ as it only scales ζ_f and ζ_m without affecting qualitative behavior. Furthermore, since F-actin polymerization generates protrusive forces at the plasma membrane and myosin activity produces contractile forces in the cortical actin network, all active forces in our model are applied exclusively to the membrane particles, representing the cell cortex. Frictional force appears between cortex and substrate and takes the form $\mathbf{f}_i = \eta\mathbf{v}_i$. The cytosolic and external fluid particles remain subject only to passive DPD interactions. Note that the overdamped condition is met naturally by choosing a large value for dissipative coefficients γ_{ij}.

H. Computational Resources

Simulations run on compute nodes equipped with Intel Xeon Platinum 8480+ (Sapphire Rapids) CPUs, with up to 112 physical cores per node (224 logical cores) and approximately 488 GiB of RAM. Typical production runs used 1 node, which composes 112 MPI ranks, and 0.4286 ± 0.0079 wall-clock hours per 10^6 integration steps for a system of 6.4×10^4 particles.

III. ANALYSIS

All simulations are conducted in standard Lennard-Jones (LJ) reduced units, in which the fundamental mass, distance, and energy are set to 1. For clarity and comparability across parameter choices, we present all results in a dimensionless form obtained by rescaling the simulation outputs as follows: time is scaled by the F-actin decay rate $\tilde{t} = \mu_f t$, and length by the initial cell radius $\tilde{l} = l/R_0$. These choices set a characteristic speed $u_0 = R_0\mu_f$. Unless stated otherwise, we report the dimensionless center-of-mass speed $\tilde{u} = u/u_0$ as the migration speed of the cell. We scale concentrations by their respective initial uniform values, $\tilde{C}_f = C_f/C_{g,0}$ for F-actin and $\tilde{C}_{m_a} = C_{m_a}/C_{m_b,0}$ for cortex-recruitable myosin, where $C_{g,0}$ and $C_{m_b,0}$ are the initial uniform concentrations of G-actin and inactive myosin, respectively. With these scalings, the dimensionless control parameters associated with actin polymerization and myosin conversion parameters can be defined as $\tilde{p} = pC_{g,0}^2/\mu_f$ and $\tilde{A}_{\max} = A_{\max}C_{m_b,0}^2/\mu_f$.

IV. MORPHODYNAMICS

To assess the model's intrinsic morphodynamics, we quantify the steady-state motility and shape as we vary the actin polymerization parameter, $\tilde{p}$, and the myosin conversion parameter $\tilde{A}_{\max}$, with stochastic pseudopod activation turned off. The cell shape is described by the aspect ratio, defined as $\chi = L/W$, where L and W are the cell length along the migration direction and width perpendicular to the migration direction, respectively. We perform 10 repetitions for each simulation condition and analyze the ensemble-averaged results. Initial conditions and simulation parameters are provided in the Appendix.

We summarize the intrinsic morphodynamics of the model in Fig. 2, in which (a) shows steady-state speed $\tilde{u}$ in the plane of the dimensionless actin polymerization rate $\tilde{p}$ and myosin conversion rate $\tilde{A}_{\max}$. The speed increases when either rate is raised and then gradually saturates at large $\tilde{p}$ and $\tilde{A}_{\max}$. Figure 2(b) shows two representative one-dimensional parameter cuts at $\tilde{p}/10^2 = 3.2$ and $\tilde{A}_{\max}/10^2 = 0.8$. At a fixed $\tilde{p}$, increasing the myosin conversion rate first speeds up the cell and then leads to a speed plateau at strong myosin activity. At fixed $\tilde{A}_{\max}$, increasing actin polymerization $\tilde{p}$ displays a similar rise-and-saturation behavior. Therefore, within the explored parameter range, both strong actin polymerization and myosin conversion promote rapid migration. Figure 2(c) illustrates the polarity of the cytoskeleton by showing the steady-state concentration fields of cortex-recruitable myosin and F-actin in a migrating cell. Cortex-recruitable myosin motors accumulate at the rear, providing contractile forces, whereas F-actin is enriched at the cell front, driving membrane protrusion. The spatial polarity and coordination between F-actin-dependent protrusion and myosin-dependent contraction in our model qualitatively align with experimental observations in amoeboid cells, such as dendritic cells [3, 6, 32]. Finally, Fig. 2(d) characterizes the relation between the steady migration speed and cell shape. The data fall into two branches depending on the relative strength of actin polymerization $\tilde{p}$ and myosin conversion $\tilde{A}_{\max}$. When $\tilde{p} \geq 4\tilde{A}_{\max}$, protrusion dominates over contractility, and the cell elongates in the migration direction. In this pulling-dominated regime, the speed $\tilde{u}$ increases with the aspect ratio χ, consistent with a previous phase-field model [10], as shown by the red curves. In contrast, contractility dominates in the regime $\tilde{p} \leq 4\tilde{A}_{\max}$ and the strong rearward pushing

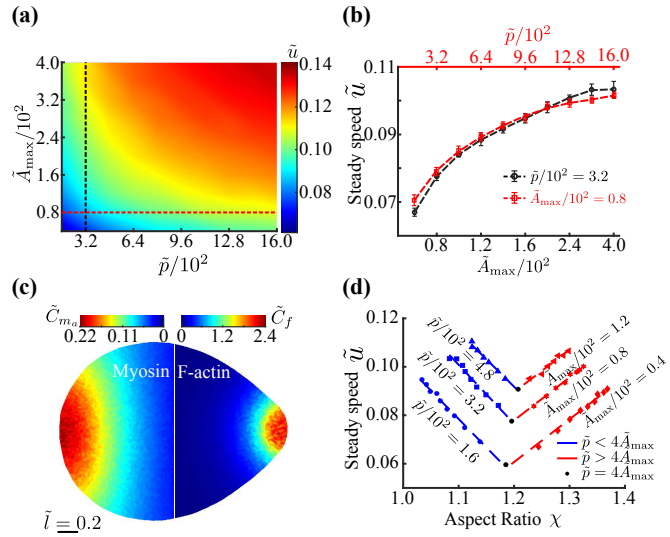

FIG. 2. Morphodynamics of the simulated cell. (a) Steady-state migration speed $\tilde{u}$ as a function of the dimensionless actin polymerization rate $\tilde{p}$ and the myosin conversion rate $\tilde{A}_{\max}$. Black and red dashed lines indicate the parameter cuts shown in (b). (b) Speed profiles along the black (vertical) and red (horizontal) dashed lines in (a), illustrating the increase and saturation of $\tilde{u}$ with both $\tilde{p}$ and $\tilde{A}_{\max}$. (c) Representative steady-state concentration fields of F-actin $\tilde{C}_f$ and cortex-recruitable myosin $\tilde{C}_{m_a}$ in a migrating cell. The scale bar shows the dimensionless length $\tilde{l} = 0.2$, where R_0 is the initial radius for the cell. (d) Correlation between the steady speed $\tilde{u}$ and the cell aspect ratio χ. Blue and red colors distinguish pushing- and pulling-dominated branches, respectively, by the relation between $\tilde{p}$ and $\tilde{A}_{\max}$.

shortens the cell length along the migration direction. In this pushing-dominated regime, increasing $\tilde{A}_{\max}$ reduces the aspect ratio χ while increasing $\tilde{u}$, resulting in the negative correlation observed in the blue curves.

V. FREE MIGRATION ON TWO-DIMENSIONAL SURFACE

We next study the migration of our model on a two-dimensional surface. We focus on the migration characteristics of various F-actin polymerization activities $\tilde{p}$ while keeping control parameters of the dynamics unchanged. We perform 20 simulations for each condition to acquire reliable statistics. The mean square displacement (MSD) curves shown in Fig. 3(b) imply that enhancing polymerization activity increases the MSD over the observation time, thus improving the two-dimensional motility of a cell. To quantitatively describe the cell motility, we use the time-dependent diffusion coefficient $\tilde{D}(\tilde{\tau})$ [33], which is defined as $\tilde{D}(\tilde{\tau}) = \frac{1}{4}\frac{\mathrm{d}\langle\Delta\tilde{x}^2\rangle}{\mathrm{d}\tilde{\tau}}$. The long-time limit of $\tilde{D}(\tilde{\tau})$ recovers the definition of the classical diffusion coefficient. Plateaus in $\tilde{D}(\tilde{\tau})$ curves at a large $\tilde{\tau}$ represent the long-time cell motility. Results shown in Fig. 3(c) confirm that cells with a larger $\tilde{p}$ exhibit a higher $\tilde{D}(\tilde{\tau})$, thereby being more motile. Next, we ask whether typical timescales are affected by the dynamics, despite the explicit stochastic processes with prescribed timescales, such as τ_{live} for pseudopods. First, we look at the velocity autocorrelation (Fig. 3(d)). We define the velocity half-correlation time $\tilde{\tau}_{\frac{1}{2}}$ as the time duration for the normalized velocity correlation to drop from 1 to 0.5. Note that $\tilde{\tau}_{\frac{1}{2}}$ is sometimes referred to as persistence time in other studies [18, 34]. The velocity autocorrelation curves shown in Fig. 3(d) almost overlap, together with the inset that displays the zoom-in region around the half-correlation area, implying minor differences between correlation times measured from various $\tilde{p}$. Based on Fig. 3(d) and (e), $\tilde{\tau}_{\frac{1}{2}}$ approximately lies in the range $13 \leq \tilde{\tau}_{\frac{1}{2}} \leq 15$, which matches the half of the $\tilde{\tau}_{\mathrm{live}} = 25$ set for the simulations, suggesting that the correlation time is hardly affected by the intracellular dynamics in our model. Since $\tilde{\tau}_{\frac{1}{2}}$ is very likely controlled by the simulation parameter $\tilde{\tau}_{\mathrm{live}}$, our model is not supposed to obtain a speed-persistence correlation [34] that is commonly observed in amoeboid cell migration. Figure 3(e) confirms that there is no linear correlation between $\tilde{u}$ and $\tilde{\tau}_{\frac{1}{2}}$. We introduce another timescale $\tilde{\tau}_{\frac{\pi}{2}}$, which we call the turning time, defined as the time needed for a cell to change its moving direction over 90 degrees to its initial direction. To extract the $\tilde{\tau}_{\frac{\pi}{2}}$, we split a trajectory into consecutive turning segments, within which the moving direction is within 90 degrees of its direction at the start of the segment. For each trajectory, we compute the mean segment duration $\tilde{\tau}_{\frac{\pi}{2}}$ and mean segment speed $\tilde{u}$ over all segments. Interestingly, we obtained a linear correlation between $\tilde{u}$ and $\tilde{\tau}_{\frac{\pi}{2}}$. This discrepancy likely results from the definition of persistence time for cell migration. In previous studies [18, 34], $\tilde{\tau}_{\frac{1}{2}}$ and $\tilde{\tau}_{\frac{\pi}{2}}$ are interchangeable since the two definitions differ only up to an irrelevant constant factor [34]. However, our model explicitly decouples the two timescales, resulting in separate dependencies on cell speed.

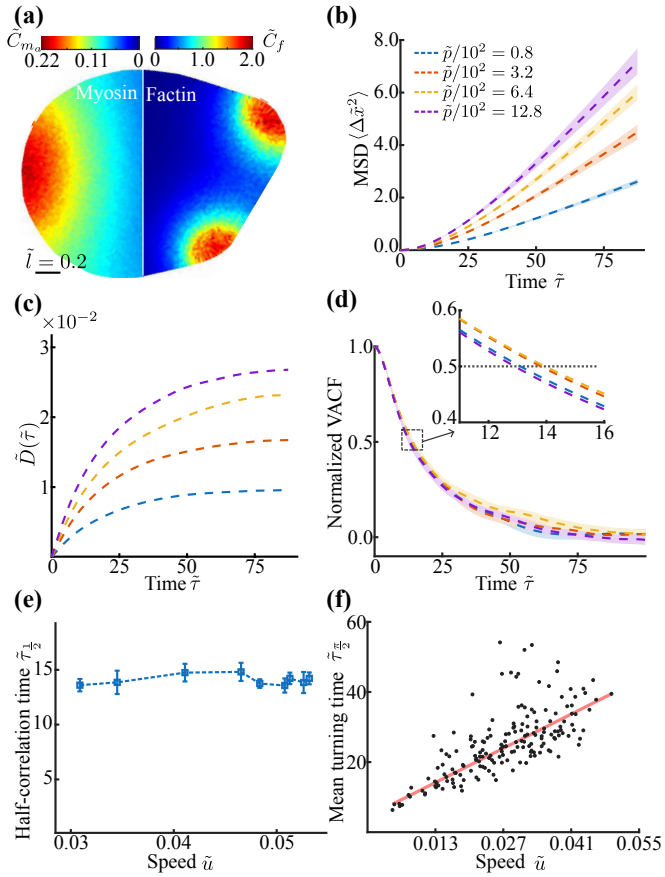

FIG. 3. Migration on a two-dimensional surface. (a) Representative concentration fields of F-actin and cortex-recruitable myosin in the cell. (b) Mean square displacement $\langle\Delta\tilde{x}^2\rangle$ at different F-actin polymerization activities $\tilde{p}$. Dashed lines show the average curve over the 20 repetitions for each $\tilde{p}$. Color shadings indicate the standard error of the mean (SEM). The legend is shared in (b)-(d). (c) Time-dependent diffusion coefficient $\tilde{D}(\tilde{\tau})$ at different active polymerization activities $\tilde{p}$. (d) Normalized velocity autocorrelation function (VACF). Dashed lines show the average VACF curves over 20 repetitions for each $\tilde{p}$. Color shading indicates the SEM. The inset displays the zoom-in area around the half-correlation region. (e) Correlation between velocity half-correlation time $\tilde{\tau}_{\frac{1}{2}}$ and speed $\tilde{u}$. $\tilde{\tau}_{\frac{1}{2}}$ is the time point when the normalized velocity autocorrelation drops to 0.5. Speed $\tilde{u}$ is the average speed of a trajectory. Error bars show the SEM. (f) Correlation between cell-mean turning time $\tilde{\tau}_{\frac{\pi}{2}}$ and speed $\tilde{u}$. A turning event is defined as a change in migration direction by more than 90° relative to the segment start direction. $\tilde{u}$ is the mean trajectory speed and $\tilde{\tau}_{\frac{\pi}{2}}$ is the mean duration of turning segments.

VI. CARGO-LOADED MIGRATION IN POROUS ENVIRONMENT

Migration through porous, confining microenvironments is a hallmark of leukocyte trafficking in tissues, where cells must traverse micrometric gaps that can be smaller than the cell diameter. To demonstrate the capability of our model in complex environments, we focus on the migration of a cargo-loaded cell in a porous environment formed by regularly spaced pillars. The spacing between pillars remains unchanged among all simulations.

Figure 4(a) shows a representative snapshot from our simulations, while Fig. 4(b) shows a representative image of a dendritic cell migrating in a pillar forest with 9-μm spacing, with myosin IIA labeled by green fluorescent protein (GFP). Although a single GFP label does not distinguish cortex-recruitable from cytosolic myosin, Fig. 4(b) indicates that myosin is distributed throughout the cell, with a clear enrichment toward the rear. Our model qualitatively recapitulates this spatial distribution. We study the migration of cells with no internal cargo, a soft cargo and a stiff cargo. The cargo stiffness is determined by the bending rigidity K_θ, and we take $K_\theta = 1$ and $K_\theta = 100$ for soft and stiff cargos, respectively, while keeping all other mechanical properties unchanged. We calculate the mean speed for individual trajectories and group them by the cargo characteristics as shown in Fig. 4(c). The result demonstrates that cargo-free cells have a mean migration speed comparable to cells with a stiff cargo, while being slower than cells with a soft cargo, suggesting a mechanically and geometrically induced change in migration. To interpret this trend, we introduce two quantities, moving speed and stall fraction. The moving speed is defined as the average speed of a cell that moves faster than a low-speed threshold, chosen as $0.3u_0$, where u_0 is the median of the mean speeds of cargo-free cells. The stall fraction is defined as the fraction of frames in which the cell moves slower than $0.3u_0$. Figure 4(d) shows that cargo-free cells are less likely to get stuck between pillars but generally move more slowly than cargo-loaded cells. Moreover, the stall fraction increases with the cargo stiffness, consistent with the intuition that a stiff cargo promotes geometric trapping. To study the cause of the speedup, we examine the mean concentration of F-actin and cortex-recruitable myosin along the cortex (Fig. 4(e)). Strikingly, cargo loading does not enhance the F-actin accumulation, but selectively increases the concentration of cortex-recruitable myosin along the cortex, revealing a cargo-induced "myosin bump-up" in the model. Cells with a soft cargo accumulate more myosin, leading to a higher moving speed. Since the cargo is soft, the cell can pass between pillars without becoming trapped. This indicates that the speedup is driven by enhanced contractility rather than increased protrusive activity. In contrast, a stiff cargo also increases myosin, but geometric trapping dominates, leading to frequent stalls.

There are experimental observations supporting the biological plausibility of cargo-induced changes in motility. First, dendritic cells infected by *Toxoplasma gondii* exhibit enhanced motility via rapid cytoskeleton remodeling [35], and recent work suggests that elevated contractility is crucial for dendritic cells carrying a parasite cargo to migrate under confinement [24]. Second, if one interprets the nucleus as a prominent intracellular cargo, differences in nuclear deformability correlate with migration speeds. For example, dendritic cells have soft and deformable nuclei, allowing them to move at a rapid speed of a few microns per minute in 3D [32], whereas fibroblasts typically have stiffer nuclei and move more slowly

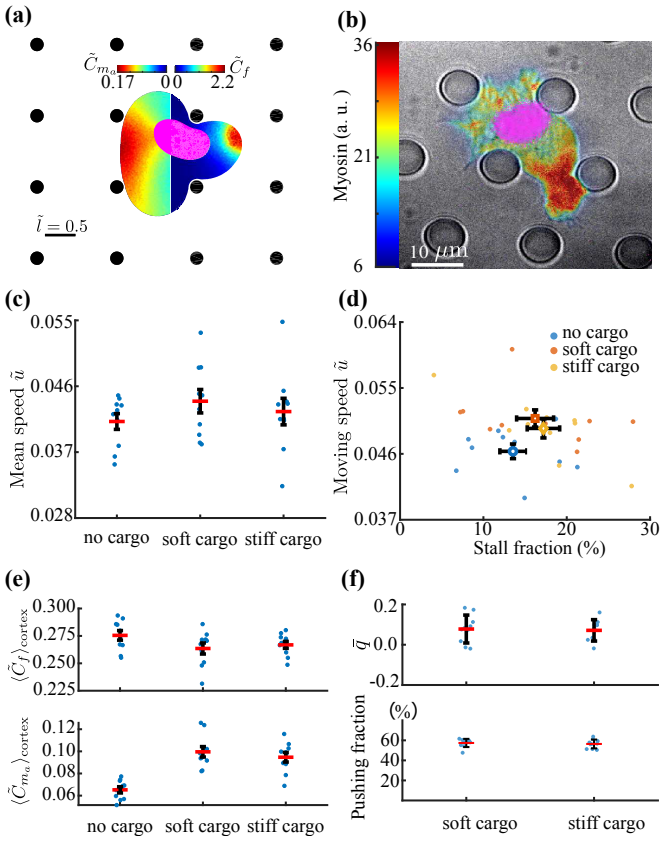

FIG. 4. Migration of cargo-loaded cells in a pillar forest. (a) Snapshot of a cargo-loaded cell in a pillar forest. (b) Representative picture of a dendritic cell moving in a pillar forest. The spacing between pillars is around 9 μm. Color shows the myosin labeled by Green fluorescent protein and is represented in arbitrary units (a.u.). The nucleus is in purple. (c). Mean speed $\tilde{u}$ statistics of cells under various cargo conditions. (d) Moving speed $\tilde{u}$ versus stall fraction under various cargo conditions. (e) Cortical concentrations of F-actin and cortex-recruitable myosin are represented by the mean concentrations along the cortical particles. (f) Locomotion of the cargo. Relative motion q is defined as the projection of the relative velocity of the cargo to the cell in the direction of the cell movement, and $\bar{q}$ is the time average of q during the entire trajectory. Therefore, $\bar{q} > 0$ indicates a forward transport of the cargo. Pushing fraction is the frame fraction for which q is positive.

than one micron per minute through the ECM [36].

Moreover, our model has the advantage of tracing cargo locomotion. To demonstrate how a cargo is transported within the cell, we quantify the relative motion by $q = (\mathbf{v}_{\text{cargo}} - \mathbf{v}_{\text{cell}}) \cdot \hat{\mathbf{v}}_{\text{cell}}$, where $\mathbf{v}_{\text{cargo}}$ and $\mathbf{v}_{\text{cell}}$ are the cargo center-of-mass velocity and cell center-of-mass velocity, respectively. $\hat{\mathbf{v}}_{\text{cell}}$ indicates the unit direction of $\mathbf{v}_{\text{cell}}$. The mean relative motion $\bar{q} = \frac{1}{T}\int_0^T q\mathrm{d}t$, where T is the total duration of a trajectory. Positive values of q imply that the cargo moves toward the cell moving direction relative to the cell motion at a given time, while $\bar{q}$ represents an overall relative motion of a trajectory. In Fig. 4(f), the statistics of $\bar{q}$ show that cargos with both stiffnesses are transported to the cell front. Since there is no mechanical coupling between the cargo and the cell, the forward cargo transport results from the pushing force of the cytosolic flow. We further compute the pushing fraction as the fraction of frames in which the cargo moves forward relative to the cell, i.e., $q > 0$. This suggests that a soft cargo is pushed to the front more frequently than a stiff cargo.

In our model, in a pillar-forest environment, cargo loading reshapes migration by jointly tuning mechanical interactions with the geometry and the cell's contractility. Soft cargo leads to faster motion with limited stalling, while stiff cargo promotes trapping despite similar mean-speed statistics to cargo-free cells. Crucially, these differences correlate with a cargo-induced increase in cortex-recruitable myosin carried by cortex particles without a corresponding increase in cortical F-actin, identifying enhanced myosin contractility as the dominant intracellular response to cargo loading in a confined environment. The model further reveals robust forward cargo transport driven by cytosolic pushing, which occurs more frequently for soft cargo.

VII. CONCLUSION

In summary, we present a mesoscopic particle-based model that combines intracellular dynamics and cellular mechanics, utilizing transport dissipative particle dynamics (tDPD) and smoothed particle hydrodynamics (SPH). In this particle-based setup, intracellular and extracellular regions are naturally distinguished by particle types, while a spring-connected membrane chain enforces a deformable, impermeable boundary that prevents fluid leakage. No-flux boundary conditions for biochemical species are implemented straightforwardly by restricting concentration exchange to particle pairs inside the cell, enabling stable and realistic intracellular regulatory dynamics. Using SPH, we map concentration fields to protrusive and contractile forces on the cortex, thereby providing a comprehensive description that links biochemical patterning, cell deformation, and migration. Our model demonstrates the capability to capture amoeboid migration across multiple perspectives, including intrinsic morphodynamics and motility in both free space and complex porous environments. Overall, the results qualitatively align with established theoretical frameworks, capture a range of experimentally observed behaviors, and provide a flexible platform for exploring how intracellular regulation and mechanical interactions jointly affect amoeboid cell migration in complex environments.

VIII. APPENDIX

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TABLE II. Essential model parameters used in this work. All quantities are in LJ reduced units, where we set fundamental mass, distance, and energy to 1.

Parameter	Symbol	Value	Description
Number of cortex particles	N_{cortex}	500	
Number of cytosol particles	N_{cytosol}	8000	
Cortex spring stiffness	K_l	100	
Cortex bond rest length	l_0	0.5	
Area elasticity	K_S	0.01	
Target area	S_0	4.97e+03	
Perimeter elasticity	K_P	1	
Target perimeter	P_0	250	
Bending rigidity	K_θ	10	
Rest angle	θ_0	$\frac{pi}{2}$	radians
Time step	Δt	0.001	time interval
Temperature	$k_B T$	0.1	
DPD cutoff	r_c	1.3	
DPD conservative coeff.	a	20	
DPD dissipative coeff.	γ	20	
F-actin mobility	κ_f	5	
G-actin mobility	κ_g	50	
Cortex-recruitable myosin mobility	κ_{m_a}	30	
Cytosolic myosin mobility	κ_{m_b}	50	
Concentration-exchange cutoff	r_{cc}	4	
Protrusion lifetime	τ_{live}	500	lifetime of a protrusion
Protrusion cooldown	τ_{cd}	43	cooldown time for a protrusion being reactivated
Protrusion birth rate	α_{birth}	0.0015	global birth rate of a new protrusion
Max protrusions	$n_{\max}$	2	maximum number of protrusions per cell
OU time constant	τ	50	Ornstein-Uhlenbeck time constant
OU noise strength	σ	20	noise strength of the membrane-wise OU process
Polymerization radius	r_{poly}	10	radius of the protrusion source region
Actin half-saturation	K_f	1	half-saturation constant for actin production
Actin Hill coefficient	n_f	2	Hill coefficient for actin production
Actin degradation rate	μ_f	0.05	degradation rate of actin
G-actin half-saturation	K_g	1	G-actin half-saturation constant
G-actin Hill coefficient	n_g	2	Hill coefficient for G-actin consumption
Myosin half-saturation	K_{m_b}	1	half-saturation constant for cytosolic myosin converting to cortex-recruitab
Cortex-recruitable myosin Hill coefficient	n_{m_a}	2	Hill coefficient for cytosolic myosin converting to cortex-recruitable my
Cortex-recruitable myosin degradation rate	μ_m	0.05	rate of cortex-recruitable myosin converting to cytosolic myosin
F-actin inhibition half-saturation	$K_{f,m}$	2e-04	effectively accumulate myosin to the cell rear
F-actin inhibition Hill coefficient	n_{m_b}	2	
Protrusion force coefficient	ζ_f	1.5	protrusion force constant
Contraction force coefficient	ζ_m	1.0	contraction force constant

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